Financial Conditions, r^*, and the Transmission of Monetary Policy Surprises

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Abstract

High-frequency monetary policy surprises identify policy shocks cleanly, but their estimated effects on unemployment and inflation in post-1994 data are often weak or puzzling. We show that two features of the monetary transmission mechanism resolve much of this. First, policy reaches the economy through financial conditions as a whole—long-term yields, equity valuations and the dollar—not through a single short-term rate. Second, the level of interest rates must be measured relative to the neutral rate r^*, which trended down for most of the period. We propagate surprises measured in 30-minute FOMC windows through a mixed-frequency model in which daily financial variables drive monthly unemployment and core inflation. With a nominal two-year yield as the only channel, the familiar puzzles appear: unemployment falls and industrial production rises after a tightening. Adding a market-based financial conditions index turns the responses conventional; measuring rates and the index relative to the Laubach–Williams r^* makes them precise and stable. A policy move that raises the two-year yield by 1 percentage point raises the unemployment gap by about 0.3 percentage point within a year and lowers the core price level by about 0.5 percent after two years—similar to, and for unemployment larger and more persistent than, the best recent estimates (Bauer and Swanson, 2023a). The two ingredients complement existing remedies based on information effects and on purging predictable surprises.

1 Introduction

The high-frequency identification of monetary policy shocks is one of the major empirical advances in macroeconomics. Changes in interest-rate futures and Treasury yields in narrow windows around FOMC announcements isolate news about policy from other news (Kuttner, 2001; Gürkaynak et al., 2005), and their effects on asset prices are estimated precisely (Bernanke and Kuttner, 2005; Nakamura and Steinsson, 2018). Using these surprises to learn about the effects of policy on output, unemployment and inflation has been harder. In samples that start in the 1990s, the estimated responses are often small, imprecise or of the wrong sign (Ramey, 2016). The literature has offered two main explanations and remedies. One is that policy announcements reveal the Federal Reserve's information about the economy, so that a surprise tightening also signals good news (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). The other is that surprises are partly predictable from public data that predate the announcement, reflecting the market's misperception of the Fed's response to the economy; orthogonalizing them restores conventional and larger responses (Bauer and Swanson, 2023a,b). A third strand stresses that the transmission of policy through financial markets must be modeled: Gertler and Karadi (2015) use the one-year yield and the excess bond premium, and Caldara and Herbst (2019) show that omitting credit spreads, to which policy reacts, attenuates all responses.

This note proposes a simpler and complementary explanation, rooted in the transmission mechanism itself. Policy affects spending through the whole configuration of financial conditions—the cost of long-term credit, the value of wealth, and the exchange rate (Boivin et al., 2010)—and what matters for spending is the level of real interest rates relative to their neutral level r^*, which fell substantially from the 1990s to the 2010s (Laubach and Williams, 2003; Holston et al., 2017; Kiley, 2020). Empirical models that carry a high-frequency surprise forward through a single nominal short-term rate neglect both features. The first omission matters because the surprise moves long yields, equities and the dollar along with the short rate, and these components carry much of the effect on activity. The second matters because, over the post-1994 period, rates trended down with r^* while activity trended up; a reduced-form link between the level of rates and activity then partly reflects these opposing trends rather than the cyclical stance of policy.

We take a direct route. The shock is the Nakamura and Steinsson (2018) policy-news surprise, measured in 30-minute windows (253 FOMC announcements, 1994–2026). Its impact on the two-year yield and, when included, on a market-based financial conditions index (FCI) is read off the window, where macroeconomic variables do not move. A daily model carries the financial variables forward from the announcement day, and monthly unemployment and core inflation respond to monthly averages of the daily variables, as in a standard monthly VAR. The model respects the timing of each announcement within its month, which addresses the temporal-aggregation problem stressed by Kilian (2024) and Chudik and Georgiadis (2022). The FCI is the market index of a companion note (FCI note, 2026): its weights on the 10-year yield, equity valuation and the real dollar are their responses to the same policy news, and it is built in three steps—nominal, real, and relative to r^*.

The evidence builds in four steps. With the nominal two-year yield as the only channel, the post-1994 puzzles appear: unemployment falls after a tightening, significantly with richer lag structures, prices do not move, and industrial production rises. Adding the FCI turns the unemployment response conventional. Measuring rates in real terms changes little. Measuring rates and the FCI relative to r^* is decisive: the unemployment response becomes significant at horizons of one to three years for every lag length, the core price level falls significantly, and industrial production falls. Neither ingredient works alone: removing r^* without the FCI leaves the puzzle in place, and the FCI without r^* gives the right sign but imprecise estimates. The magnitudes are close to the benchmark of Bauer and Swanson (2023a): unemployment rises about 0.3 percentage point in the first year against their 0.2, and stays higher for longer; the price level falls by about 0.5 percent after two years, as in their estimates. We obtain these results from FOMC announcements alone, without orthogonalizing the surprises.

These results complement rather than compete with the information-effect and predictability literatures. Our propagation is estimated from the reduced-form link between financial conditions and the economy, so it is less exposed to the correlation between surprises and prior news that biases proxy VARs (Bauer and Swanson, 2023a), and the surprise reaches macro variables only through financial conditions, which screens out information channels that bypass markets. Both mechanisms may therefore be at work alongside ours. Our point is that two standard features of the transmission mechanism—financial conditions as a whole, and the level of rates relative to r^*—go a long way toward resolving the puzzles on their own, and that they have been absent from the high-frequency-identification literature.[1]

[1] This statement is based on the papers we have read, including those cited above; we have not found high-frequency studies that propagate the shock through a broad financial conditions index or measure the policy rate relative to r^*. The closest is Caldara and Herbst (2019), who add a credit spread.

2 Data and method

Surprises and window responses. The shock is the Nakamura–Steinsson policy-news factor from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025): the first principal component of 30-minute changes in five short-term interest-rate futures around 253 FOMC announcements, February 1994 to September 2026. We scale it so that the two-year Treasury yield rises 1 percentage point in the window. The window response of the FCI (0.99 per percentage point) is computed from the 30-minute changes in the 10-year yield, the S&P 500 and the dollar index in the same database, combined with the FCI's weights. Unemployment and prices are zero in the window.

Financial conditions. The FCI is the market index of FCI note (2026): a weighted sum of the 10-year Treasury yield, equity valuation (cyclically adjusted price–earnings ratio, CAPE) and the real broad dollar, with weights equal to their responses to the same policy news, so that a unit of the index is the effect of a policy move that raises the two-year yield 1 percentage point. It comes in three versions, built in the same steps as in that note. *Nominal*: the variables as measured. *Real*: the 10-year yield net of 10-year expected inflation. r^**-gap* (baseline): the real 10-year yield net of the one-sided Laubach–Williams r^* (New York Fed, current estimates), and equities as the gap between the earnings yield and r^*. The two-year yield follows the same rule: nominal; net of two-year expected inflation (Cleveland Fed; Haubrich et al., 2012); and net of expected inflation and r^*. Daily versions follow the daily 10-year yield, CAPE and dollar and are shifted, smoothly within each month, so that their monthly averages equal the monthly indexes exactly; the adjustments for expected inflation and r^* move slowly, so the window responses are the same in every version.

Model. The model has a daily and a monthly block. In the daily block, the two-year yield and (when included) the FCI follow a heterogeneous autoregression in their 1-, 5-, 22- and 66-day averages and respond to the current and previous month's unemployment gap and core inflation (estimated 1990–2026). In the monthly block, the unemployment gap (the CBO-based gap of the companion note) and core PCE inflation are rows of a monthly VAR with L lags of each and L lags of the monthly averages of the daily financial variables, estimated by OLS over February 1994–September 2026 with March–December 2020 excluded; L is 3, 6 or 12. Within each month macro conditions are set first, and markets respond to them; in the announcement month, macro variables have not yet moved. The response to a surprise is computed by placing the window impact on the announcement day (drawn from the empirical distribution of FOMC dates within the month), simulating the daily block forward, averaging within months, and iterating the monthly block. This is a restricted mixed-frequency VAR in the sense of Ghysels (2016), with equal weights within the month; the restriction is the one implied by a standard monthly VAR, and we could not reject it for unemployment.

Identifying assumption. The surprise affects unemployment and prices only through the variables in the daily block, and the economy responds to a given change in those variables in the same way whatever caused it. The first part is weaker the more complete the set of financial variables; the second is the assumption behind any use of financial conditions indexes to summarize the stance of policy (Hatzius et al., 2010; Ajello et al., 2023). A direct regression of unemployment on the surprises that does not impose it (Chudik and Georgiadis, 2022; Chudik and Kilian, 2026) is about ten times less precise and cannot discriminate between our estimates and zero.

Inference. We resample announcements (which re-estimates the window impact and the distribution of announcement dates) and residuals of the monthly equations, 200 draws. We report 68 and 90 percent intervals. Recent high-frequency studies report 90 percent bands (Bauer and Swanson, 2023a; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016), Gertler and Karadi (2015) 95 percent, and Jarociński and Karadi (2020) both 68 and 90 percent.

3 Results

Table 1 and Table 2 report the responses of the unemployment gap and the core PCE price level at 12, 24 and 36 months, for 3, 6 and 12 monthly lags, at each step. Figure 1 shows the four key cases.

Table 1: Unemployment gap (percentage points), per 1 percentage point policy-induced rise in the two-year yield

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	-0.13^{*}	-0.27^{**}	-0.34^{**}	-0.01	-0.19^{*}	-0.36^{**}	0.08	-0.08	-0.25^{*}
Nominal	yes	0.18^{**}	0.09	0.07	0.38^{**}	0.29^{**}	0.19^{*}	0.42^{**}	0.37^{**}	0.24^{**}
Real	no	-0.02	-0.14^{*}	-0.25^{**}	0.07	-0.10	-0.28^{**}	0.12	-0.03	-0.21^{*}
Real	yes	0.25^{**}	0.18^{*}	0.12	0.40^{**}	0.33^{**}	0.24^{**}	0.38^{**}	0.36^{**}	0.26^{**}
r^{*}-gap	no	0.08	0.03	-0.08	0.16^{*}	0.04	-0.15^{*}	0.20^{*}	0.05	-0.16
r^{*}-gap	yes	0.30^{**}	0.32^{**}	0.34^{**}	0.38^{**}	0.39^{**}	0.41^{**}	0.34^{**}	0.38^{**}	0.37^{**}

Columns: horizon and number of monthly lags. **: 90 percent interval excludes zero; *: 68 percent interval excludes zero. Rates and FCI at each step as described in Section 2. Bauer and Swanson (2023a), benchmark: +0.19 at 12 months (90 percent band excludes zero), +0.17 and +0.15 at 24 and 36 months (band includes zero), read from their Figure 8.

Table 2: Core PCE price level (percent), per 1 percentage point policy-induced rise in the two-year yield

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	0.01	0.06	0.01	-0.08	0.03	0.05	-0.17	-0.07	0.03
Nominal	yes	-0.09	-0.07	-0.10	-0.28^{*}	-0.24^{*}	-0.23	-0.42^{*}	-0.41^{*}	-0.36^{*}
Real	no	-0.10^{*}	-0.02	-0.07	-0.19^{*}	-0.08	-0.06	-0.26^{*}	-0.16	-0.08
Real	yes	-0.17^{*}	-0.14^{*}	-0.18^{**}	-0.33^{*}	-0.32^{*}	-0.31^{*}	-0.44^{*}	-0.47^{**}	-0.41^{*}
r^{*}-gap	no	-0.15^{*}	-0.13^{*}	-0.19^{**}	-0.22^{*}	-0.26^{*}	-0.31^{*}	-0.28^{*}	-0.39	-0.44^{*}
r^{*}-gap	yes	-0.18^{*}	-0.25^{**}	-0.33^{**}	-0.28^{*}	-0.48^{**}	-0.61^{**}	-0.37^{*}	-0.68^{**}	-0.84^{**}

See Table 1. Bauer and Swanson (2023a), benchmark (CPI): −0.36, −0.48 and −0.6 at 12, 24 and 36 months, 90 percent bands exclude zero.

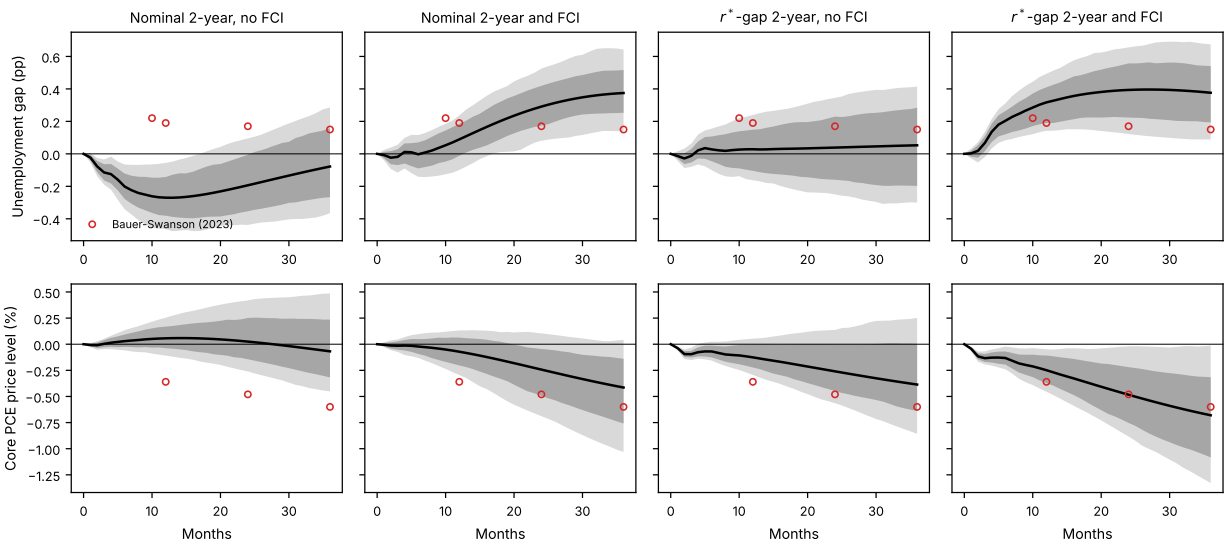


Figure 1: Responses to a policy move that raises the two-year yield 1 percentage point (6 lags). Dark band: 68 percent; light band: 90 percent. Circles: Bauer and Swanson (2023a), read from their Figure 8.

Step 1: the standard set-up reproduces the puzzles. With the nominal two-year yield as the only channel—the policy indicator of Bauer and Swanson (2023a)—unemployment falls after a tightening, significantly with 6 and 12 lags, and the price level does not move. This is the post-1994 pattern documented by Ramey (2016) and the one Bauer and Swanson (2023a) obtain with unadjusted surprises.

Step 2: financial conditions. Adding the FCI turns the unemployment response conventional at every horizon and lag length. It is significant at 36 months, but at 12 months it is weak with 6 and 12 lags, and the price level falls without being significant at conventional levels. At the 68 percent level, which some studies report (Jarociński and Karadi, 2020), the price-level decline is significant at 36 months for every lag length and at 24 months with 3 and 6 lags: with the FCI there is no price puzzle.

Step 3: real rates. Measuring the two-year yield and the FCI net of expected inflation changes little, as in the companion note, where the nominal and real indexes are close after 1990: expected inflation moved little in this period.

Step 4: rates relative to r^*. Measuring the two-year yield and the FCI relative to r^* is decisive, but only together. Without the FCI, the r^*-gap rate leaves unemployment insignificant. With it, unemployment rises by 0.30–0.34 percentage point at 12 months and 0.34–0.38 at 36 months, significant at the 90 percent level at every horizon and lag length and nearly identical across lag lengths. The price level falls by 0.48–0.61 percent at 24 months with 6 and 12 lags, significant at the 90 percent level. At the 68 percent level the price decline is significant at every horizon and lag length. Removing r^* also helps without the FCI, but only modestly: the price level then falls (significant at the 68 percent level at most horizons), while unemployment stays insignificant.

Comparison with Bauer and Swanson (2023a). Their benchmark proxy VAR (1973–2020, with surprises from FOMC announcements and Chair speeches, orthogonalized to pre-announcement news) gives a peak unemployment response of about 0.2 percentage point after 10 months and a CPI decline of about 0.5 percent after two years, per percentage point on the two-year yield. Our step-4 estimates match the price response and give a larger and more persistent

unemployment response, with intervals that exclude zero through 36 months where theirs include zero after about a year and a half. Their estimates use 30 years of instrument data, Chair speeches and orthogonalization; ours use FOMC windows alone and no orthogonalization.

4 Robustness

Industrial production. Adding industrial production to the monthly block (Table 3) leaves the pattern for unemployment and prices unchanged. Without the FCI, industrial production rises after a tightening at every step; with the FCI it falls; at the r^* step it falls about 0.9 percent at 12 months, significantly with 6 and 12 lags, against about 1.5 percent in Bauer and Swanson (2023a), and recovers faster.

Table 3: Industrial production (percent), monthly block with industrial production added

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	1.15**	1.15**	1.64**	1.05*	1.10*	1.63**	1.01*	1.04*	1.23*
Nominal	yes	−0.17	−0.36	−0.08	−0.10	−0.14	−0.09	0.50	0.35	0.27
Real	no	0.70*	0.68*	1.13**	0.71*	0.83*	1.29**	0.76	0.92*	1.04*
Real	yes	−0.36	−0.59*	−0.36	−0.12	−0.20	−0.16	0.48	0.30	0.27
r^*-gap	no	0.22	0.15	0.43	0.26	0.56	0.87	0.33	0.82	0.77
r^*-gap	yes	−0.47*	−0.88**	−0.91**	−0.14	−0.32	−0.51	0.36	0.10	−0.12

See Table 1. Bauer and Swanson (2023a): about −1.5, −1.0 and −0.85 at 12, 24 and 36 months, read from their Figure 8.

Which trend? Table 4 replaces the Laubach–Williams estimate with Kiley (2020)'s, and replaces the real and r^* adjustments with two statistical trends: a quadratic time trend in every monthly equation, and the two-year yield measured relative to its trailing 8-year average. Three results stand out. First, the FCI is essential in every variant: without it, unemployment is insignificant or significantly negative. Second, with the FCI, a statistical trend removes the unemployment puzzle as well as r^* does: with a quadratic trend, unemployment rises 0.25–0.44 percentage point at 12 months and 0.6–0.8 at 24–36 months, significant at the 90 percent level in all but one case, and with the 8-year average (nominal or real FCI), 0.15–0.25 at 12 months and 0.27–0.38 at 24–36 months. Third, the price level is where the Laubach–Williams gap matters. With the quadratic trend and the FCI, the price level falls by 0.3–0.5 percent at 24–36 months—no price puzzle, and significant at the 68 percent level at 24–36 months with 6 lags (and with 12 lags for the nominal FCI)—but not at the 90 percent level; with the 8-year average it barely moves; with Kiley's r^*, unemployment is significant at the 90 percent level with 3 lags and at 12 months with 6 lags, but weak with 12 lags, and the price level is significant only at 12 months at the 68 percent level. The 8-year average and the Laubach–Williams trend are close until 2015 and diverge after 2016, when the 8-year average of the two-year yield fell to about 1 percent while expected inflation plus r^* stayed near 3–4 percent. The evidence therefore supports a general point—the slow-moving component of rates must be removed—and a specific one: the Laubach–Williams gap, which embeds a model of the inflation-relevant neutral rate, gives the most precise price responses. Using r^* estimates as a measure of the stance in high-frequency studies raises the usual concerns about their uncertainty (Laubach and Williams, 2003; Holston et al., 2017; Kiley, 2020), which our intervals do not include.

Table 4: Alternative trends: unemployment gap (top) and core PCE price level (bottom)

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
r^*-gap (LW), baseline	yes	0.30**	0.32**	0.34**	0.38**	0.39**	0.41**	0.34**	0.38**	0.37**
r^*-gap (Kiley)	no	0.00	−0.11	−0.22**	0.06	−0.10	−0.29**	0.09	−0.07	−0.25*
r^*-gap (Kiley)	yes	0.38**	0.28**	0.15*	0.47**	0.36*	0.14	0.37**	0.30*	0.09
Nominal + quadratic trend	no	0.07	−0.13	−0.27**	0.36*	0.10	−0.19	0.53**	0.29	−0.00
Nominal + quadratic trend	yes	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
Nominal + quadratic trend	real	0.41**	0.32**	0.25*	0.69**	0.67**	0.57**	0.68**	0.74**	0.70**
2-year − 8-year average	no	−0.22**	−0.30**	−0.36**	−0.23*	−0.28*	−0.46**	−0.22	−0.21	−0.40**
2-year − 8-year average	nominal	0.23**	0.18*	0.15*	0.38**	0.36**	0.27*	0.34**	0.36**	0.27**
2-year − 8-year average	real	0.25**	0.19**	0.15*	0.37**	0.36**	0.28*	0.31**	0.34**	0.29*
2-year − 8-year average	r^*-gap	0.06	0.08	0.14*	0.10*	0.15*	0.18*	0.08	0.15*	0.18

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
r^*-gap (LW), baseline	yes	−0.18*	−0.25**	−0.33**	−0.28*	−0.48**	−0.61**	−0.37*	−0.68**	−0.84**
r^*-gap (Kiley)	no	−0.13*	−0.07	−0.12*	−0.18	−0.11	−0.11	−0.22	−0.16	−0.10
r^*-gap (Kiley)	yes	−0.13*	−0.12	−0.16*	−0.21	−0.15	−0.06	−0.29	−0.21	0.08
Nominal + quadratic trend	no	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26*	−0.34	−0.18
Nominal + quadratic trend	yes	−0.09	−0.11	−0.11	−0.24	−0.36*	−0.31*	−0.33	−0.54*	−0.49*
Nominal + quadratic trend	real	−0.09	−0.11	−0.13	−0.23	−0.34*	−0.33	−0.31	−0.51*	−0.50
2-year − 8-year average	no	0.19*	0.10	0.02	0.37*	0.22	0.10	0.50*	0.28	0.17
2-year − 8-year average	nominal	0.01	−0.07	−0.11	−0.02	−0.19	−0.23	−0.02	−0.24	−0.31
2-year − 8-year average	real	0.02	−0.07	−0.14*	0.02	−0.15	−0.28*	0.05	−0.19	−0.38
2-year − 8-year average	r^*-gap	0.07	−0.05	−0.15*	0.19	−0.07	−0.27*	0.29	−0.06	−0.35

See Table 1. FCI column: the FCI version used (no FCI; nominal; real; r^*-gap). The 8-year average and Laubach–Williams trends (two-year expected inflation plus r^*) are close until 2015 (averages 5.3 and 5.5 percent over 1994–2007; 2.3 and 2.5 over 2008–2015) and diverge after 2016 (1.4 and 3.6).

5 Conclusion

High-frequency surprises identify monetary policy shocks well; the difficulty has been in tracing their effects on the economy. We show that two standard features of the transmission mechanism go a long way toward resolving the puzzles in post-1994 data. Policy works through financial conditions as a whole, and it is the level of rates relative to the neutral rate that matters for spending and inflation. A model that carries window-identified surprises through a broad financial conditions index, measured with rates relative to r^*, delivers conventional, precise and lag-robust responses of unemployment and prices, of a size comparable to the best recent estimates, without additional instruments or purging. Neither ingredient works alone.

These findings complement the information-effect and predictability literatures rather than replacing them; the three mechanisms may operate together, and combining orthogonalized surprises (Bauer and Swanson, 2023a) with our propagation is a natural next step. Three limitations deserve emphasis. The identifying assumption—that policy reaches the economy through the financial variables in the model, and that the economy responds to them the same way whatever moved

them—is strong and cannot be tested precisely with direct regressions. The results for prices depend on the Laubach–Williams estimate of r^*; statistical trends remove the unemployment puzzle but give weaker price responses. And credit spreads, which Caldara and Herbst (2019) show matter for both transmission and the policy rule, are not in our window impact. The broader lesson is that the decline in r^* over the past three decades is not only a fact about the level of rates but also a measurement problem for estimating the effects of policy.

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